

Machine Learning for Flight Delay Prediction

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Abstract—Flight delays are a widespread and costly issue affecting millions of passengers each year, leading to schedule disruptions and billions of dollars in economic losses. This project presents a data science pipeline to predict flight arrival delays by integrating airline operations data, historical weather observations, and airport metadata. Focusing on the top 10 busiest U.S. airports between December 2022 and March 2023, we processed and combined data from the Bureau of Transportation Statistics (BTS), the National Oceanic and Atmospheric Administration (NOAA), and OpenFlights. With extensive feature engineering, including cyclical time encoding, traffic intensity, and weather metrics, we evaluated multiple machine learning models. Our results indicate that flight delays exhibit complex, non-linear dependencies between weather and scheduling, which are effectively captured by the Random Forest classifier, outperforming Logistic Regression with 74.00% accuracy, a 7× increase in recall, and an ROC-AUC of 0.71.

Index Terms—Data Science, Machine Learning, Flight Delays, Predictive Modeling, Random Forest

I. INTRODUCTION

Flight delays represent a widespread AND costly challenge within the modern aviation industry, affecting millions of passengers annually. Beyond the immediate inconvenience to travelers, these delays trigger cascading schedule disruptions across flight networks, resulting in billions of dollars in economic losses. Airlines face inflated operational costs, compromised aircraft rotations, and crew scheduling conflicts, while passengers suffer significant losses in both time and capital. The inherent operational complexity of high-traffic airports, combined with unpredictable atmospheric conditions, makes mitigating these delays a critical priority for transportation stakeholders.

Accurately forecasting flight disruptions holds immense value. By anticipating delays before they propagate through the network, airlines can optimize schedules ahead of time, allocate resources more efficiently, and mitigate financial losses. Furthermore, reliable predictions empower passengers to make informed travel decisions, ultimately improving the overall travel experience.

The objective of this project is to develop a data science pipeline capable of predicting flight arrival delays using real-world datasets. Specifically, we aim to predict whether a flight will be delayed by more than 15 minutes, capturing the complex relationship between weather events and airport scheduling. To achieve this, our methodology leverages a

subset of aviation and atmospheric data spanning December 2022 through March 2023.

By combining Airline On-Time Performance records from the Bureau of Transportation Statistics (BTS), historical weather observations from the National Oceanic and Atmospheric Administration (NOAA), and geographic metadata from OpenFlights, we establish a comprehensive feature set for the top ten busiest airports in the United States. Through extensive data cleaning, feature engineering, and predictive modeling, we evaluate the efficiency of two machine learning classifications. Specifically, we compare Logistic Regression and Random Forest architectures, identifying conditions highly susceptible to flight delays.

II. RELATED WORK

The prediction of flight delays using machine learning has been explored extensively, primarily driven by the availability of large aviation datasets and the economic incentives for solving this problem. Prior research consistently highlights the efficiency of methods and the necessity of integrating external factors, such as weather, to capture the underlying causes of schedule disruptions.

Several studies have established baseline comparisons between various machine learning algorithms for binary delay classification. For instance, recent evaluations of classifiers on flight datasets have demonstrated that Random Forest and Decision Tree models consistently outperform linear models like Logistic Regression and Support Vector Machines, particularly when coupled with resampling techniques to handle the inherent class imbalance of flight delays [1]. These tree based models perform very well at capturing the complex relationships between flight schedules and delay occurrences.

Further research has specifically targeted the integration of atmospheric conditions with operational data. A 2024 IEEE study focused on predicting flight delays by combining aviation metrics with weather constraints [2]. Evaluating models such as Ridge, Lasso, and Random Forest, the authors concluded that Random Forest achieved the highest performance, highlighting its performance in processing more diverse data sets. Similarly, other frameworks have merged Bureau of Transportation Statistics (BTS) records with National Oceanic and Atmospheric Administration (NOAA) weather data, supporting our claim that temporal variables (e.g., departure hour,

day of the week) and atmospheric signals are crucial for accurate prediction of flight delays [3].

A. Distinction of Our Approach

While our approach builds upon the much explored success of combining BTS and NOAA datasets using Random Forest classifiers, it is unique from previous works in scope and feature representation. Many existing studies attempt to generalize across the entire U.S. airspace or focus on a single regional airport. In contrast, our approach isolates the top 10 busiest U.S. airports during the winter travel season (December 2022 – March 2023). This targeted scope allows our models to specifically learn the dynamics of highly congested hubs during periods of severe weather volatility.

Furthermore, we extend traditional temporal feature engineering. Rather than treating time as a simple categorical or linear variable, we implemented cyclical time encoding (using sine and cosine) to preserve the continuous, looping nature of daily flight schedules. Finally, we use a dynamic traffic intensity feature to quantify the specific volume of flights per airport, directly mapping operational congestion to delay probability in a way that standard flight data had not captured.

III. DATA DESCRIPTION

To accurately model the conditions that lead to flight delays, our analysis required a large amount of both flight data and historical meteorological conditions.

A. Data Sources

Our predictive pipeline leverages data from three distinct, publicly available sources:

- **Commercial Airline Data:** Our primary dataset, sourced from the Bureau of Transportation Statistics (BTS), titled Airline On-Time Performance records. This dataset includes large amounts of operational metrics such as carrier information, origin and destination, scheduled versus actual departure/arrival times, and cancellation status.
- **Weather Data:** Atmospheric conditions were obtained using the National Oceanic and Atmospheric Administration (NOAA) Integrated Surface Database. We extracted hourly observations from weather stations located near our target airports, focusing on high-impact variables including temperature, dew point, wind speed and direction, visibility, and atmospheric station pressure.
- **Airport Metadata:** To ensure accurate mapping and data merging, we utilized the OpenFlights database. This provided essential geographic features, including precise airport latitude/longitude coordinates, altitude, and time-zone information.

B. Data Scoping and Reduction

The raw, combined datasets presented several significant challenges. The historical weather observations alone constitute very large amounts of data globally. To ensure computational feasibility within the scope of this project, and to avoid overanalyzing data, we strictly constrained our parameters.

Our analysis was restricted to a four month winter travel window, spanning December 2022 through March 2023. Furthermore, due to the excessive size of the weather data, we filtered our geographic scope exclusively to the top 10 busiest U.S. airports by flight origin. This targeted scoping was highly effective, reducing our initial pool of approximately 2.3 million raw flight records down to a refined, and still highly relevant dataset of around 700,000 flights.

C. Exploratory Data Analysis

Initial exploratory analysis of the roughly 700,000 filtered flights revealed several significant insights into delay distributions. We defined our primary target variable as an arrival delay exceeding 15 minutes. Summary statistics indicated a pronounced class imbalance, this is typical of aviation data, where the majority of flights arrive on time, but a minority experience substantial delays driven by some compounding and uncontrollable external factors.

IV. METHODOLOGY

To transform our raw flight and atmospheric datasets into a reliable predictive model, we implemented a comprehensive data science pipeline. This approach was divided into three primary phases: general data cleaning, feature engineering, and model development. The following subsections detail the techniques applied at each stage to maximize our predictive performance.

A. Data Cleaning and Integration

The data cleaning and integration pipeline was implemented in Python using the `pandas` library. Given the scale and number of unique features in the three data sources, careful preprocessing was needed to handle any missing values and align the datasets temporally.

Flight Data Preprocessing: The initial flight dataset was filtered to include only flights originating from the ten target U.S. airports (ATL, ORD, DEN, DFW, CLT, PHX, SEA, LAX, LAS, LGA). Because our objective is to predict arrival delays for flights that actually complete their routes, all cancelled flights were removed from the dataset. Rows missing critical target variables, specifically departure and arrival delays, were dropped. To prevent data leakage during the modeling phase, columns containing any post-departure information like actual departure time, actual arrival time, and taxi times were intentionally excluded. Missing values in the `WeatherDelay` feature were imputed with zero, assuming no weather related operational delay was recorded that day.

Weather Data Preprocessing: The NOAA weather data required significant standardization. Weather station identifiers (USAF codes) were mapped directly to the corresponding IATA airport codes. Because meteorological data is continuous and, in our case, often recorded with missing hourly intervals, we applied a grouped linear interpolation strategy. Continuous variables, such as dry bulb temperature, dew point, wind speed, and station pressure, were interpolated based on the specific airport's timeline to preserve same day trends. Any remaining

missing values after interpolation were filled using the median of their respective columns. Categorical and anomalous entries were also standardized. For instance, trace amounts of precipitation (recorded as "T") were converted to numeric zeros, and missing sky condition entries were categorized as "UNKNOWN".

Temporal Alignment and Merging: The most significant challenge involved merging the flight data with the atmospheric observations. Flight departure schedules (CRSDepTime) are stored as integer blocks (e.g., 1430 for 2:30 PM). These integers were parsed, combined with the FlightDate, and rounded down to the nearest hour (e.g., 14:00) to create a DEP_HOUR timestamp.

Similarly, the weather observation timestamps were floor-rounded to the nearest hour. To handle instances where multiple weather observations were recorded within a single hour, the weather dataset was sorted chronologically and stripped of duplicates, containing only the first observation per hour per airport. Finally, a left join was executed to merge the cleaned weather data into the flight records, matching on the origin airport code and the floored departure hour. A left join was chosen over an inner or right join to ensure that no valid flight records were dropped in the event of missing meteorological observations. Ultimately, this ensured every flight was paired with the atmospheric conditions present at its scheduled time of departure while preserving the full scope of the data.

B. Feature Engineering

To maximize the predictive power of our models, we engineered several features from the raw data, capturing temporal patterns, operational congestion, and meteorological severity.

Target Variable Definition: Our primary objective is binary classification. We defined the target variable, IsDelayed, by thresholding the ArrDelay column. Any flight arriving more than 15 minutes after its scheduled arrival time was classified as delayed (1), while all others were classified as on time (0).

Temporal and Cyclical Encoding: Standard temporal features like month, day of the week, and departure hour were extracted from the scheduled departure. Because time is cyclical (e.g., hour 23 is immediately followed by hour 0), treating hours as a linear feature can confuse machine learning algorithms. To preserve this continuous, looping nature, we applied a transformation to the departure hour, creating two new features: $\sin(2\pi \cdot \text{Hour}/24)$ and $\cos(2\pi \cdot \text{Hour}/24)$.

Operational Traffic Intensity: We initially thought that airport congestion would be a leading indicator of delays. To quantify this, we engineered an AirportTraffic metric by calculating the total number of flights scheduled to depart from a specific origin airport within the same hour. Additionally, we created an AirlineAvgDelay feature, representing the historical mean departure delay associated with each specific marketing airline network.

Meteorological Interactions: Beyond raw weather data, we engineered features to capture severe conditions. We created a binary IsFreezing flag triggered when the dry bulb temperature dropped to or below 32°F. We also constructed

a WindVisibilityInteraction feature by multiplying hourly wind speed by the inverse of visibility (with a small constant added to avoid division by zero), capturing the increased risk associated with strong winds under low visibility conditions.

C. Modeling and Evaluation Strategy

With our data cleaned and features constructed, the dataset was prepared for machine learning application. To ensure a strong evaluation and to prevent data leakage, we implemented a preprocessing and splitting strategy using the scikit-learn library.

Data Preprocessing and Splitting: The dataset was split into training and testing sets utilizing an 80/20 ratio. Because flight delays are inherently imbalanced (the majority of flights are on-time), we applied stratified sampling during the split to ensure the distribution of the IsDelayed class remained consistent across both sets. A ColumnTransformer pipeline was utilized to scale and encode the features independently. Numeric features went through median imputation for missing values followed by standardization via StandardScaler. Categorical features (Origin, Destination, and Airline Network) were imputed with the most frequent value and transformed using One-Hot Encoding.

Model Architectures: We trained two distinct models to evaluate performance:

- **Logistic Regression (Baseline):** We established a linear baseline using Logistic Regression. Due to the high dimensionality introduced by the one-hot encoded categorical variables, the maximum iterations were increased to 5000 to ensure convergence.
- **Random Forest Classifier:** To capture the complex data more accurately, for example, non-linear interactions between weather severity and airport scheduling, we implemented a Random Forest architecture. The model was configured with 50 estimators and a maximum depth of 25 to balance complexity and overfitting. Crucially, to mitigate the class imbalance favoring on-time flights, we utilized the `class_weight='balanced_subsample'` parameter, which adjusts weights inversely proportional to class frequencies based on the bootstrap sample for every tree.

Evaluation Metrics: Model performance was evaluated using multiple metrics appropriate for such classification tasks. Beyond standard Accuracy, we calculated Precision, Recall, and the F1-Score to specifically measure the models' effectiveness at identifying delays. Furthermore, we calculated the Receiver Operating Characteristic Area Under the Curve (ROC-AUC) and generated confusion matrices to visually interpret classification behavior and the trade-offs between false positives and missed predictions. Finally, we extracted importance metrics from the Random Forest to identify the most significant predictive features.

V. RESULTS

The performance of both the baseline Logistic Regression and the Random Forest models were evaluated on the test set. Given the pronounced class imbalance inherent in aviation data, standard accuracy is an insufficient and often misleading metric. Therefore, our analysis heavily prioritized Precision, Recall, the F1-Score, and the Receiver Operating Characteristic Area Under the Curve (ROC-AUC). The complete performance comparison is detailed in Table I.

A. Comparative Model Performance

The Logistic Regression baseline achieved a high accuracy of 77.34%. However, a deeper analysis reveals the model's severe limitations. The model yielded a recall of only 6.25%. In practice, this indicates that the linear model overwhelmingly defaulted to predicting the majority class (on-time flights), failing to detect delays under more minimal condition changes. The resulting F1 Score of 0.11 confirms its overall poor performance for this task.

On the other hand, the Random Forest classifier demonstrated a superior capability in identifying true delays. While the overall accuracy experienced a minor reduction to 74.00%, this tradeoff proved highly beneficial for the minority class detection. The Random Forest achieved a recall of 46.01%. This is greater than a 7x increase over the baseline. This means the Random Forest successfully identified nearly half of all actual delayed flights. This massive improvement in sensitivity drove the F1 Score up to 0.45, representing a 4x improvement over the linear baseline.

B. Trade-offs and Classification Behavior

While the Random Forest excelled in recall, it resulted in a drop in precision from 62.37% (Logistic Regression) to 44.27%. This indicates a higher rate of false positives, where the model predicts a delay that does not ultimately manifest. However, in the context of airline operations and passenger routing, a false positive (preparing resources for a delay that is mitigated) may generally be less economically damaging than a false negative (being caught completely unprepared by an unexpected disruption).

Furthermore, the Random Forest demonstrated a lot better overall separability between the delayed and on time classes, achieving a higher ROC-AUC of 0.71 compared to the baseline's 0.65. Visualized in Figure ??, the confusion matrices further illustrate this classification behavior, clearly displaying the Random Forest's superiority in mapping the more complex, atmospheric and operational conditions that lead to flight disruptions.

VI. DISCUSSION

The results of our analysis highlight the necessity for stronger modeling techniques in aviation data. The following sections interpret the performance between our baseline and Random Forest models, while also acknowledging the constraints of our dataset and project scope.

TABLE I
MODEL PERFORMANCE COMPARISON

Metric	Logistic Regression	Random Forest
Accuracy	77.34%	74.00%
Precision	62.37%	44.27%
Recall	6.25%	46.01%
F1 Score	0.11	0.45
ROC-AUC	0.65	0.71

A. Model Interpretation

The contrast in recall and F1-Score between the Logistic Regression baseline and the Random Forest classifier uncover the complex nature of aviation delays. These delays are rarely driven by a single isolated factor(s). Instead, disruptions occur from the more complex, intersection of multiple variables.

Logistic Regression assumes a linear relationship between the independent variables. Consequently, it struggled to define accurate decisions for majority of the not obvious delays, ultimately defaulting to the majority "on-time" class. For example, a minor increase in wind speed might not affect a flight on a clear afternoon, but that same wind speed combined with freezing temperatures and high hourly airport traffic will almost certainly increase the odds of a delay.

The Random Forest architecture naturally excels in this environment. By utilizing a group of deep decision trees (maximum depth of 25 in this study), the model was able to partition the feature space, capturing thresholds where weather severity and airport congestion intersect. Furthermore, the Random Forest was able to leverage our engineered features, such as the cyclical time encoding and the WindPressure metric, identifying multi-dimensional patterns that the linear baseline failed to capture.

B. Limitations

While our pipeline successfully demonstrates the predictive power of aviation and atmospheric data, several limitations restrict a stronger predictive ability.

First, the temporal resolution of our weather data introduces a notable constraint. By relying on hourly atmospheric summaries, the model assumes weather conditions stay for a full 60-minute window. In reality, atmospheric events can be sudden or brief periods of intense weather. These events can disrupt flight schedules in a matter of minutes. Our model is essentially blind to these sudden fluctuations.

Second, to maintain project feasibility, our project scope was restricted strictly to the top 10 busiest U.S. airports. While this successfully captured the dynamics of major hub congestion, it limits the model's applicability. Less popular regional airports possess vastly different operational strategies, runway throughput, and weather susceptibilities. A model trained exclusively on hubs like ATL and ORD may not generalize well to smaller airfields or different localized conditions.

Finally, the dataset lacks visibility into internal airline logistics. A significant portion of delays are cascading where a single delay might trigger a chain reaction across multiple

airports and flights. Without data regarding specific aircraft turnaround times, crew scheduling constraints, and precise delay patterns, our model can only infer these bottlenecks through our engineered features like `AirlineAvgDelay` and `AirportTraffic`.

VII. CONCLUSION & FUTURE WORK

This project successfully demonstrated the viability of a data science pipeline to predict flight arrival delays. By integrating operational flight logs with atmospheric observations, we engineered a feature set and trained models that captured the realities of high traffic U.S. airports. Our evaluation shows that a Random Forest classifier can effectively model the relationship between weather and scheduling congestion. Using weather statistics, temporal features, and traffic intensity metrics, it significantly outperforms a standard linear baseline in identifying true delays.

While these initial results are promising, several paths exist for future work. To improve predictive performance, future iterations should incorporate higher-resolution atmospheric inputs, such as 15-minute interval data, rather than relying on hourly summaries. Furthermore, the pipeline could be expanded from a binary classification task into a regression framework to predict the exact duration of anticipated delays. Finally, integrating deeper operational metrics would provide the model with the necessary context to predict delays across the broader aviation network. These metrics include historical airport congestion trends, route specific delay patterns, and individual aircraft turnaround times.

VIII. INDIVIDUAL CONTRIBUTIONS

The successful completion of this project required a collaborative effort, with tasks divided to leverage each team member's strengths. A detailed breakdown of specific contributions and personal reflections from each author are provided below.

TABLE II
TEAM MEMBER CONTRIBUTIONS

Member	Contributions
Aiden Slabiak	Data Engineering & Feature Engineering: Led cleaning, preprocessing, dataset filtering, and implemented original features.
Nathan Hubbell	Modeling & Deployment: Led development, training, tuning of predictive models, and generated the final dashboard/report visualizations.

Personal Reflection: Nathan Hubbell

Personal Reflection: Aiden Slabiak

Throughout this project, I gained a deeper understanding of how important data engineering and feature construction are to the success of a machine learning pipeline. The most challenging part of my work was managing and integrating such large and complex datasets. In particular, aligning the hourly NOAA weather observations with BTS flight departure times and reducing the dataset from 2.3 million raw records

to a clean set of 700,000 flights required careful processing and organization.

If I were to approach this project again, I would focus on building a more scalable data pipeline, potentially using cloud computing resources. This would allow us to process a higher volume of weather data (e.g., 15-minute intervals) and expand the analysis beyond the top 10 busiest airports.

REFERENCES

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APPENDIX A: AI USAGE LOG